Practice Problems
Calculus Physics II

Name: _____ **Date:** _____

Chapter 51 –Applications of Electromagnetic Induction

1) Bob swings a sheet of copper in pendulum fashion between the poles of a strong magnet. The sheet slows in entering the magnetic field because it

Eddy Currents

(A) induces swirling currents in the copper and accompanying magnetic fields that resist motion.
B) experiences both air and magnetic friction.
C) is repelled by free electrons in the copper.
D) none of the above

2) Which of the following devise use eddy currents to its advantage?

A) Electric guitar
B) Traffic signal
C) Wireless charger
(D) Magnetic brakes

3) If you drop a bar magnet in a vertical copper pipe it will fall slowly because

A) of air resistance.
(B) it induces a magnetic field in the pipe that resists motion of the magnet.
C) the copper is a good conductor of both electricity and magnetism.
D) of electron repulsion.

4) A solid block of aluminum (m = 3 kg) is sliding on a frictionless surface with some initial velocity, it encounters a magnetic field and comes to a stop in 0.8 s, the electrical power created by the eddy currents was 4 W, what was the initial velocity of the block? (Assume no loss of energy transferring between electrical and mechanical)

$\Delta E = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -\frac{1}{2}mv_i^2$

$Power = P = \frac{\Delta E}{\Delta t} = \frac{\frac{1}{2}mv_i^2}{\Delta t}$ | Solve For v_i | $v_i = \sqrt{\frac{2P\Delta t}{m}} = 1.46\ m/s$

5) Which cooking pot would NOT work on an Induction stove?

(A) Glass pot
B) Iron pot
C) Aluminum pot
D) Stainless steel pot
E) All the above

6) What make a GFI plug safe?

(A) It detects if there is no return current to the plug.
B) It detects a change in voltage.
C) It limits the current the plug can transmit.
D) It limits the about of voltage a plug can have across its terminals.

7) Which best describes the principle behind how a speaker works?

A) When a ferromagnetic material is brought near a coil wrapped around a magnet there will be an induced voltage.
B) When a coil of wire has a changing current, it will induce a current in a second coil of wire.
C) The current moving through a coil of wire will experience a force if that coil is in a magnetic field.
D) If there is a change in the magnetic field that a coil of wire experiences, then there will be an induced current.
E) None of the above.

8) Which best describes the principle behind how a car traffic light sensor works?

A) When a ferromagnetic material is brought near a coil wrapped around a magnet there will be an induced voltage.
B) When a coil of wire has a changing current, it will induce a current in a second coil of wire.
C) The current moving through a coil of wire will experience a force if that coil is in a magnetic field.
D) If there is a change in the magnetic field that a coil of wire experiences, then there will be an induced current.
E) None of the above.

9) Which best describes the principle behind how an electric guitar works?

A) When a ferromagnetic material is brought near a coil wrapped around a magnet there will be an induced voltage.
B) When a coil of wire has a changing current, it will induce a current in a second coil of wire.
C) The current moving through a coil of wire will experience a force if that coil is in a magnetic field.
D) If there is a change in the magnetic field that a coil of wire experiences, then there will be an induced current.
E) None of the above.

10) Which best describes the principle behind how a microphone works?

A) When a ferromagnetic material is brought near a coil wrapped around a magnet there will be an induced voltage.
B) When a coil of wire has a changing current, it will induce a current in a second coil of wire.
C) The current moving through a coil of wire will experience a force if that coil is in a magnetic field.
D) If there is a change in the magnetic field that a coil of wire experiences, then there will be an induced current.
E) None of the above.

11) Which best describes the principle behind how an induction charger works?

A) When a ferromagnetic material is brought near a coil wrapped around a magnet there will be an induced voltage.
B) When a coil of wire has a changing current, it will induce a current in a second coil of wire.
C) The current moving through a coil of wire will experience a force if that coil is in a magnetic field.
D) If there is a change in the magnetic field that a coil of wire experiences, then there will be an induced current.
E) None of the above.

12) An electromagnetic flowmeter is useful when it is desirable not to interrupt the system in which the fluid is flowing (such as the blood in an artery during heart surgery). Such a device is illustrated in the figure. The conducting fluid moves with speed v in a tube of diameter d. Perpendicular to this tube is a magnetic field B. A voltage V is induced between opposite sides of the tube due to the motion of the conducting fluid in the magnetic field. For a certain case, $B = 0.120$ T, $d = 1.2$ cm, and the measured voltage is $V = 2.88$ mV. Determine the speed of the fluid.

$|\vec{B}| = 0.12\,T$

$L = d = 1.2 \times 10^{-2}\,m$

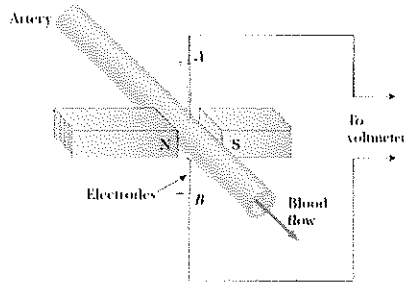
$V_{ms} = 2.88 \times 10^{-3}\,V$

$\theta = 90°$

$V_{ms} = v B L \sin\theta$

$v = \dfrac{V_{ms}}{BL\sin\theta}$

$v = 2\,m/s$



13) Faraday's law underlies the operation of

A) an electric guitar.
B) RFID chip.
C) the changing of traffic lights.
D) Wireless charging
E) all of the above
F) none of the above

14) If a RDIF chip needs 2 W of power to operate and the chip has a coil with A = 2 x 10^{-5} m^2 and N = 5 coils of wire. If the resistance in the circuit is R = 1.5 Ω, how fast must a magnetic field go from 0.5 T to 0 T in order to run the chip at 2 W?

$P = 2W$

$R = 1.5\,\Omega$

$P = \dfrac{V^2}{R}$

$V = \sqrt{PR} = 1.7V$

$V_{ms} = V = 1.7V$

$V_{ms} = -N\dfrac{\Delta\Phi}{\Delta t}$

$\Delta\Phi = A(\Phi_f - \Phi_i) = 1 \times 10^{-5}\,Wb$

→ solve for Δt

$\Delta t = \dfrac{N\Delta\Phi}{V_{ms}} = \dfrac{(5)(1 \times 10^{-5}\,Wb)}{1.7V}$

$\Delta t = 2.9 \times 10^{-5}\,s$

15) Sunspots are

A) Hotter than the Sun's surface.
B) Same temperature as the Sun's surface.
C) Cooler than the Sun's surface.

16) Why are solar flares in the shape of loops?

A) The particles are following the magnetic fields created by the sunspots.
B) The particles are following the electric fields created by the sunspots.
C) The particles are pulled back to the Sun by its large gravitational field.

KEY

1) A
2) D
3) B
4) 1.46 m/s
5) A
6) A
7) C
8) D
9) A
10) D
11) B
12) 2 m/s
13) E
14) 2.9×10^{-5} s
15) C
16) A